

Zhouhao Zhang

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Education

The Hong Kong University of Science and Technology
PhD Student, Department of Electronic and Computer Engineering
Beihang University
Bachelor of Engineering in Automation (GPA: 3.80 / 4.00)

Sep 2026 – Present
Clear Water Bay, Hong Kong
Sep 2020 – Jun 2024
Beijing

Internship

King Abdullah University of Science and Technology
Research Intern, Supervised by Prof. Shinkyu Park

Apr 2025 – Sep 2025
Jeddah, Saudi Arabia

- Developed an octree-based **coverage path planning** algorithm using ray tracing and combinatorial optimization for underwater robotic coral reef monitoring; built and validated the simulation environment in OceanSim.
- Built a multi-microdrone experimental platform, trained **reinforcement learning** controllers in IsaacGym, and deployed them with a Qualisys motion-capture system; reduced the **sim-to-real gap** through system identification.

Cytoderm
Robotics Engineer

Dec 2024 – Mar 2025
Beijing

- Developed unmanned intelligent production lines for IMU with JAKA AGV and Rokae robotic arms.
- Designed **visual servoing and force control algorithms** for robotic manipulators in precision assembly tasks.

Zhipu AI X-lab
Robotics Research Intern, Supervised by Dr. Tong Yang

Jun 2024 – Dec 2024
Beijing

- Developed the **Scene Iconicity Graph** for enhancing robotic operations, utilizing foundation models' zero-shot capabilities to improve scene understanding and reasoning.
- Integrated and deployed a **3D LiDAR SLAM** system and **vision-language navigation** on the Unitree Go2 quadruped robot, utilizing behavior trees to manage navigation tasks.
- Leading the design of a **mobile research robot** equipped with multiple sensors and AI computing resources for advanced **embodied-AI experimentation**.

Skyforce Technology
R&D Engineer Intern

Jul 2023 – Dec 2023
Shenzhen

- Calibrated structured light systems through **Gray code** and **local homography** for enhanced .
- Developed an automatic keystone correction algorithm incorporating **TOF** and **IMU**.
- Implemented obstacle avoidance for projectors using **LiDAR** and the **RANSAC algorithm**.

Skills

Languages: Chinese (Native), English (TOEFL: 98)
Programming Languages: Python, C, C++
Robot Development Environments: Linux, ROS, Gazebo, PyBullet, IsaacSim, Mujoco
Development Libraries: Pytorch, OpenCV, OpenAI Gym, PCL, Eigen
Robot Hardware: UAV, Wheeled/Quadruped Chassis, Robotic Arms, LiDAR, Depth Camera, IMU, TOF
Embedded Systems: NVIDIA Jetson, Raspberry Pi, STM32, ESP32

Honors

National Scholarship <i>National 0.2%, The Ministry of Education of the People's Republic of China</i>	Dec 2022
First Prize, 23rd China University Robot Competition (ROBOCON) <i>National Third Place</i>	Jul 2024
Second Prize, 23rd China University Robot Competition (RoboMaster)	Apr 2024
Second Prize, 22nd China University Robot Competition (ROBOCON)	Jul 2023
First Prize, 2022 Five Provinces of North China University Robot Competition	Nov 2022
First Prize, 2022 Beijing University Robot Competition	Nov 2022
Second Prize, China Intelligent Robot Fighting and Gaming Competition	Mar 2023
Third Prize, 38th National College Physics Competition	Dec 2021

Outstanding Graduate Award <i>Beihang University</i>	Jun 2024
CATIC Scholarship <i>Total 10 Places (0.3%) in Beihang University</i>	Apr 2024
First Prize, Scholarship in Discipline Competition <i>Beihang University</i>	Apr 2024
Second Prize, Scholarship in Outstanding Social Work <i>Beihang University</i>	Apr 2024
Outstanding Student Leader <i>Beihang University</i>	Nov 2023
Top Prize, Learning Excellence Scholarship <i>Top 2%, Beihang University</i>	Dec 2022
Outstanding Student <i>Top 1%, Beihang University</i>	Sep 2022
Third Prize, 32nd FengRu Cup Competition & Yuyuan Robots Competition <i>Beihang University</i>	Jun 2022

Projects

CKSRI, The Hong Kong University of Science and Technology <i>Ph.D Student</i>	Sep 2026 – Present
A visuotactile hierarchical Bayesian search method for locating and grasping buried objects in granular media.	
MedAIR, Chinese University of Hong Kong <i>Graduation Project (Remote), Guided by Prof. Qi Dou</i>	Dec 2023 – Jun 2024
- Task planning of 7DOF da Vinci Surgical Robot via demonstration-guided reinforcement learning and policy chain. - Surgical robot visual manipulation policy learning using world model-based reinforcement learning. - High-level task planning of robot based on large language models and behavior tree.	
Biomechanics and Soft Robotics Lab, Beihang University <i>Research Member, Guided by Dr. Lei Li</i>	Apr 2023 – Oct 2023
- An Aerial–Aquatic Hitchhiking Robot with Remora-Inspired Tactile Sensors and Thrust Vectoring Units, in Advanced Intelligent Systems, DOI: 10.1002/aisy.202300381. - Designed flight control programs and deployed SLAM algorithms for aerial–aquatic UAV.	
Pattern Recognition & Image Processing Course Project, Beihang University <i>Project Leader</i>	Dec 2023
- Implemented deep learning pipelines for medical image segmentation, point cloud classification and segmentation, and motor imagery classification using U-Net, PointNet/PointNet++, and EEGNet. - Developed computer vision methods for image stitching and robotic grasp detection using SIFT and Vision Transformers. - Implemented classical machine learning and search algorithms, including Gaussian mixture models with EM and graph search for combinatorial problems.	
BR Robotics Team, Beihang University <i>Member & Vice Captain</i>	Aug 2022 – Jun 2024
Visual auto-shoot algorithm and robot SE(2) state estimation for racing robot in ROBOCON 2023. Decision making algorithm with minimax search and finite state machine for racing robot in ROBOCON 2024. - Deployment of SLAM algorithms including Vins-Mono, Vins-Fusion, RTABMAP, ORB-SLAM, and FAST-LIO2. Camera pose estimation using the PNP algorithm and morphological operation. Technical guidance for the RoboMaster, ROBOCON and RoboCup competitions. - Founded Beihang University's first RoboMaster team.	

Extracurricular Experience

Red Bird Challenge Camp <i>The Hong Kong University of Science and Technology (Guangzhou)</i>	Jun 2024 Guangzhou
- Designed and built a factory simulation environment in Gazebo to simulate intelligent mobile detection robots. - Implemented 2D SLAM for mobile robots using the Cartographer algorithm. - Developed autonomous navigation and dynamic obstacle avoidance systems using the ROS Navigation stack.	
Artificial Intelligence & Machine Learning Program <i>National University of Singapore</i>	Aug 2023 Singapore
- Developed a Seq2Seq model using LSTM for accurate population forecasting. - Achieved winning team in the final presentation, earning commendation from <u>Prof. Mehul Motani</u>.	